A Homogeneous Compensation Mechanism for Indirect Evolutionary Rescue in a Roy-Style Predator–Prey Model

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Abstract

Indirect evolutionary rescue can occur when adaptive change in one species prevents extinction of another. We analyze this problem in a Roy-style predator–prey model where prey defense evolves and predator mortality stress is imposed externally. Comparing the well-mixed eco-evolutionary ODE with a spatial reaction–diffusion PDE shows that the supported mechanism is a homogeneous compensation branch rather than spatial-pattern-mediated rescue. Along this branch, increasing predator mortality stress is compensated by a decrease in the equilibrium prey defense frequency $q^*(s)$, which increases predator conversion opportunity and preserves a positive predator equilibrium over an interior stress interval. A frozen-q counterfactual, initialized at the pre-stress branch state, verifies a finite stress window, approximately $0.0726 \leq s \leq 0.1656$ on the tested grid, in which fixed defense leads to predator extinction while evolving defense preserves predators. For the linear defense trade-off ODE, the branch follows from explicit feasibility conditions and is locally stable at the tested stresses by Routh–Hurwitz inequalities. In the spatial PDE, the same homogeneous branch is linearly stable over the tested Neumann-mode range and a sampled continuous-λ modal scan over the corresponding interval. Targeted finite-amplitude heterogeneous perturbation cases that changed finite-horizon basin labels were followed to longer horizons and did not support persistent spatial-pattern-mediated rescue in the tested setup. Controlled nonlinear trade-off extensions recover the linear branch and support compensation for selected concave, convex, and mixed trade-off shapes, while remaining local and parameter-sensitive. The final interpretation is that indirect evolutionary rescue in this tested model is driven by homogeneous reaction-level compensation over the verified counterfactual rescue window.

Keywords: indirect evolutionary rescue; eco-evolutionary dynamics; predator–prey model; reaction–diffusion PDE; trade-off; Routh–Hurwitz stability

1 Introduction

Evolutionary rescue describes persistence after environmental deterioration through adaptive change [Gomulkiewicz and Holt, 1995, Bell and Gonzalez, 2009, Bell, 2017]. In predator–prey systems, rescue can also be indirect: evolution in prey may alter predator persistence. In the setting studied here, prey defense reduces predation but carries ecological costs. When predator mortality stress increases, predator pressure declines, selection can shift prey toward lower defense, and the lower-defense prey state can increase predator conversion opportunity. This feedback is the qualitative mechanism of indirect evolutionary rescue described by Yamamichi and Miner [2015].

Prey defense evolution is a natural target for this question because the defense trait changes both prey ecological performance and predator energetic gain. Rapid eco-evolutionary feedbacks in predator–prey systems can alter ecological dynamics on short timescales [Yoshida et al., 2003].

The exact form of the defense trade-off is also important: nonlinear or asymmetric trade-offs can change predator–prey eco-evolutionary dynamics [Kasada et al., 2014]. In this manuscript, trade-off geometry is therefore part of the mechanism rather than a background detail.

Spatial structure is a plausible competing explanation because local dispersal and heterogeneous predator–prey overlap can change persistence outcomes. A scalar persistence threshold is an attractive first diagnostic: one compares the mortality stress at which predators persist or fail in the ODE and PDE. That threshold framing can fail, however, when long transients, continuation dependence, or multiple reachable outcomes occur at the same stress. In that case, the correct object is not a single threshold but the mechanism organizing the reachable basins.

The mechanism supported by the present analysis is not spatial-pattern-mediated rescue. Fixed-defense spatial patterning did not produce robust predator rescue in this model sequence. The well-mixed eco-evolutionary ODE supports indirect evolutionary rescue over the frozen-q counterfactual window identified below. Subsequent ODE–PDE comparison showed that the basin structure observed in the PDE is primarily inherited from homogeneous reaction dynamics rather than generated by persistent spatial patterning. The important mechanism is therefore a homogeneous compensation branch, with spatial analyses used to test whether the PDE destabilizes or qualitatively alters that branch.

This manuscript analyzes the mechanism organizing predator persistence in the tested model. We derive the compensation branch, state its feasibility conditions, evaluate local stability using Routh–Hurwitz criteria, test spatial-mode stability in the PDE, evaluate targeted finite-amplitude heterogeneous perturbations, and examine controlled nonlinear trade-off extensions. The claim hierarchy is intentionally conservative. Strong claims are restricted to the tested ODE branch, its feasibility and local stability, the tested PDE spatial-mode stability calculation, targeted non-homogeneous perturbation outcomes, and the controlled nonlinear shape grid. Broader claims about global stability, all trade-off forms, all diffusion coefficients, or all heterogeneous initial conditions are not made.

We first formulate the ODE and PDE models. We then derive the homogeneous compensation branch and its feasibility conditions. We analyze local stability using Routh-Hurwitz criteria, test spatial-mode stability in the PDE, evaluate targeted non-homogeneous perturbations, and finally examine controlled nonlinear trade-off extensions.

2 Model

The model is a reduced Roy-style free-space eco-evolutionary formulation adapted from defended and undefended prey–predator reaction–diffusion models with free space and aposematic defense trade-offs [Sasmal et al., 2019, Roy et al., 2026]. Let n be prey density, w predator density, q prey defense frequency, and

$$z = \kappa^{-1} - n - w$$

be free space. The variable s denotes added predator mortality stress. For the linear trade-off model,

$$r(q) = r_u(1-q) + r_v q, \quad a(q) = a_u(1-q) + a_v q, \quad b(q) = b_u(1-q) + b_v q.$$

Here $r(q)$ is the prey growth trade-off, $a(q)$ is the attack or palatability trade-off, and $b(q)$ is the predator conversion trade-off. The well-mixed ODE is

$$\frac{dn}{dt} = n[r(q)z - \xi - a(q)w], \tag{1}$$
$$\frac{dw}{dt} = w[b(q)nz - (m+s) - \mu w], \tag{2}$$
$$\frac{dq}{dt} = \nu q(1-q)[(r_v - r_u)z - (a_v - a_u)w]. \tag{3}$$

Table 1 lists the values used in the reported calculations.

The spatial PDE uses the same local reaction terms:

$$\partial_t n = D_n \nabla^2 n + n[r(q)z - \xi - a(q)w], \tag{4}$$
$$\partial_t w = D_w \nabla^2 w + w[b(q)nz - (m+s) - \mu w], \tag{5}$$
$$\partial_t q = D_q \nabla^2 q + \nu q(1-q)[(r_v - r_u)z - (a_v - a_u)w]. \tag{6}$$

The PDE tests use zero-flux Neumann boundaries on a rectangular domain of side lengths L_x and L_y. The q-diffusion term is a phenomenological composition-smoothing closure for the defense-frequency variable. It is not derived here from separate defended and undefended prey density equations, and that modeling choice could affect PDE spatial-stability conclusions. The homogeneous ODE compensation branch and frozen-q counterfactual do not depend on D_q, because all diffusion terms vanish on spatially constant states.

For the controlled nonlinear trade-off extension, endpoint values are preserved but linear interpolation is replaced by shape functions:

$$r(q) = r_u + (r_v - r_u)f_r(q), \quad a(q) = a_u + (a_v - a_u)f_a(q), \quad b(q) = b_u + (b_v - b_u)f_b(q),$$

with $f_\gamma(q) = q^\gamma$. This nonlinear form is used only for controlled extension tests, not to replace the baseline linear model.

3 Methods and numerical protocols

The analysis combines targeted numerical integration with analytic or semi-analytic calculations. All numerical values reported in this manuscript are taken from CSV outputs in the repository. Additional analyses reported here are limited to ODE no-evolution counterfactual integrations and linear algebraic modal scans; they do not introduce new PDE simulations or model-equation changes.

ODE analysis. ODE trajectories were integrated from specified initial means for representative trajectories and q_0–w_0 basin maps. ODE integrations used `scipy.integrate.solve_ivp` with the LSODA method unless otherwise noted. For the no-evolution counterfactual, the horizon was $T = 3000$, with 1201 saved time points, relative tolerance 10^{-8}, absolute tolerance 10^{-10}, extinction threshold 10^{-4}, and a terminal tail window equal to the final 25 percent of the trajectory. Persistent, extinct, and transient labels were assigned from late-time predator density, terminal behavior of $q(t)$, relative change between late windows, and residual-style steady-state diagnostics. Predator-persistent outcomes have positive predator density with sufficiently small terminal change. Extinct outcomes have predator density near zero. Transient outcomes do not settle clearly into either class within the tested horizon.

Table 1: Parameter values used in the compensation-branch and spatial-stability analyses. These values define the tested mechanistic case study; they are not empirical estimates.

Symbol	Meaning	Value used	Role in model
r_u	Growth contribution of undefended prey	1	Baseline prey growth endpoint
r_v	Growth contribution of defended prey	0.65	Defense-growth trade-off endpoint
a_u	Attack or palatability endpoint for undefended prey	1	Baseline predation endpoint
a_v	Attack or palatability endpoint for defended prey	0.35	Defense-palatability trade-off endpoint
b_u	Conversion endpoint for undefended prey	0.08	Predator conversion opportunity at low defense
b_v	Conversion endpoint for defended prey	0.02	Predator conversion opportunity at high defense
κ	Inverse habitat capacity scale	0.15	Sets free-space relation $z = \kappa^{-1} - n - w$
ξ	Prey loss parameter	0.55	Appears in prey balance
m	Baseline predator mortality	0.1	Predator mortality before added stress
μ	Predator density-dependent loss	0.2	Predator self-limitation term
ν	Defense evolution rate	0.05	Scales q-dynamics
D_n	Prey diffusion coefficient	0.01	PDE prey movement
D_w	Predator diffusion coefficient	0.01	PDE predator movement
D_q	Defense-frequency diffusion coefficient	0.005	PDE defense-frequency smoothing
L_x	Domain length in x	20	Neumann spatial-mode domain scale
L_y	Domain length in y	20	Neumann spatial-mode domain scale

Compensation branch computation. For the linear trade-off model, the compensation branch was derived analytically from the interior equilibrium equations, giving closed-form z^*, w^*, n^*, and $q^*(s)$. Numerical equilibria from multiple initial guesses were used as consistency checks against the analytic branch. For nonlinear trade-offs, the branch was parameterized by q through $c(q) = r'(q)/a'(q)$ and the corresponding stress map $s(q)$.

No-evolution counterfactual. To verify that predator persistence depends on prey evolution rather than only on the existence of a positive branch, we compared the full evolving-q ODE with a frozen-q counterfactual. Both treatments were initialized at the pre-stress branch state $(n^*, w^*, q^*(0))$, with $q^*(0) \approx 0.6726$. In the counterfactual treatment $dq/dt = 0$, so q remained fixed at this pre-stress value. Stress was swept across the interior branch interval, and predator persistence/extinction was classified using the same tail-window logic as the ODE analyses. The verified rescue window is the finite-grid, finite-horizon interval where frozen-q trajectories are extinct while evolving-q trajectories persist. As a complementary analytic check, for fixed $q_0 = q^*(0)$, the prey-only frozen-q equilibrium satisfies

$$z_0 = \frac{\xi}{r(q_0)}, \qquad n_0 = \kappa^{-1} - z_0,$$

and the predator invasion boundary is

$$s_{\text{fixed}} = b(q_0) n_0 z_0 - m.$$

For the current parameterization, $s_{\text{fixed}} \approx 0.0696$, above which predators cannot invade the prey-only state when defense is fixed. In the tested finite-horizon trajectories initialized from the pre-stress branch state, predator loss occurs just above this threshold.

Local stability. Local stability of the linear branch was evaluated with the analytic reaction Jacobian. For the cubic characteristic polynomial $\lambda^3 + A_1\lambda^2 + A_2\lambda + A_3 = 0$, the Routh–Hurwitz inequalities $A_1 > 0$, $A_2 > 0$, $A_3 > 0$, and $A_1 A_2 > A_3$ determined local stability. Finite-difference Jacobians were used for selected nonlinear trade-off extensions where closed-form expressions were not the main object.

PDE spatial stability. Homogeneous ODE equilibria are homogeneous PDE steady states because diffusion vanishes on spatially constant fields. Spatial-mode stability was evaluated from $J_F(U^*) - \lambda_{ij} D$, where $J_F(U^*)$ is the reaction Jacobian, $D = \text{diag}(D_n, D_w, D_q)$, and λ_{ij} are Neumann eigenvalues on the rectangular domain. The zero mode recovers ODE stability; nonzero modes test infinitesimal spatial perturbations. Discrete Neumann modes were evaluated for $i, j = 0, \ldots, 64$, and a continuous λ-scan sampled 600 equally spaced values over the corresponding mode-equivalent λ-range.

Non-homogeneous PDE perturbations. Targeted finite-amplitude PDE tests used localized predator patches, localized defense patches, sinusoidal modes, random heterogeneity, and basin-boundary heterogeneity. PDE tests used 64×64 grids on 20×20 domains with $D_n = 0.01$, $D_w = 0.01$, $D_q = 0.005$, and explicit timestep $dt = 0.1$, with CFL checks in the simulation helper. Perturbation amplitudes were targeted rather than exhaustive: local predator patches used $A = 0.5$, local defense patches used $A_q = 0.2$, sinusoidal and random heterogeneity used $A = A_q = 0.1$, and basin-boundary heterogeneity used $A = A_q = 0.25$. Random seeds included 20260702 and 20260703 where stochastic heterogeneity was used. Each heterogeneous run was compared with a matched homogeneous control at the same stress and baseline mean state. Basin-changing heterogeneous cases were followed to longer horizons to distinguish persistent basin changes from finite-horizon transient labels.

Nonlinear trade-off extension. The nonlinear extension used the endpoint-preserving shape family $f_\gamma(q) = q^\gamma$ with $\gamma \in \{0.5, 1, 2\}$ for the growth, attack, and conversion trade-offs. This is a controlled local shape grid, not a broad scan over all nonlinear trade-off functions.

4 Homogeneous compensation branch

Define

$$\Delta r = r_v - r_u, \quad \Delta a = a_v - a_u, \quad \Delta b = b_v - b_u.$$

For an interior equilibrium with $n > 0$, $w > 0$, and $0 < q < 1$, the nontrivial equilibrium conditions are

$$r(q)z - \xi - a(q)w = 0,$$

$$b(q)nz - (m + s) - \mu w = 0,$$

$$\Delta r z - \Delta a w = 0.$$

If

$$c = \frac{\Delta r}{\Delta a},$$

then the selection-gradient condition gives

$$w^* = cz^*.$$

Because the trade-offs are linear, $r(q) - ca(q) = r_u - ca_u$, so the prey-balance equation gives

$$z^* = \frac{\xi}{r_u - ca_u}, \quad w^* = cz^*, \quad n^* = \kappa^{-1} - z^* - w^*.$$

The predator-balance equation then determines the defense frequency required at stress s:

$$q^*(s) = \frac{(m + s + \mu w^*)/(n^* z^*) - b_u}{b_v - b_u}.$$

This branch explains rescue as compensation: increasing predator mortality stress is offset by decreasing equilibrium prey defense frequency, which increases predator conversion opportunity and keeps a positive predator equilibrium feasible. In the current parameterization, the branch has approximately $n^* \approx 4.8333$ and $w^* \approx 0.6417$, while $q^*(s)$ decreases with stress.

No-evolution counterfactual. The counterfactual test starts at the pre-stress branch state and compares the full evolving-q dynamics with a frozen-q treatment in which $q = q^*(0)$ is held fixed. The frozen-q prey-only invasion calculation gives $s_{\text{fixed}} \approx 0.0696$, above which predators cannot invade the prey-only state when defense is fixed. In the tested finite-horizon trajectories initialized from the pre-stress branch state, predator loss occurs just above this threshold. On the finite targeted stress grid and integration horizon, the verified rescue window is approximately

$$s \in [0.0726, 0.1656],$$

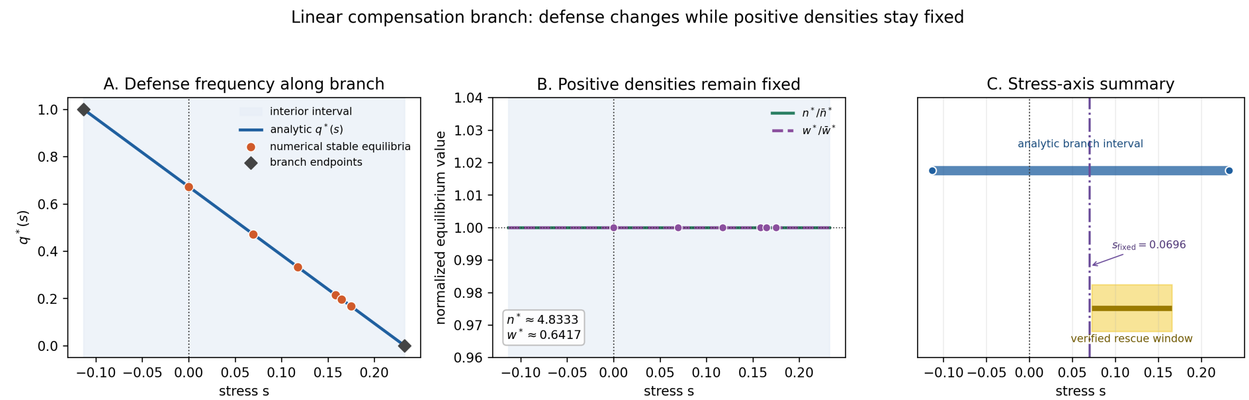

Figure 1: Compensation branch and stress-axis summary. Panel A plots analytic $q^*(s)$ against numerical stable persistent equilibria. Panel B shows normalized n^* and w^*, which remain effectively fixed along the linear branch. Panel C places the analytic branch interval, the frozen-q invasion threshold s_{fixed}, and the verified finite-grid rescue window on a common stress axis.

where frozen-q trajectories are classified as predator-extinct while evolving-q trajectories remain predator-persistent. At the representative stress $s = 0.1191$, the evolving-q trajectory lowers defense and maintains $w \approx 0.6417$, whereas the frozen-q trajectory loses the predator. This supports the indirect evolutionary rescue interpretation over this verified counterfactual window. The window is a reachable-window result from the pre-stress branch initial condition, not the full analytic feasibility interval of the interior branch. Outside this window, especially near the upper branch boundary, the pre-stress initial condition can fall outside the persistent basin even though the analytic branch remains feasible. The supplemental q_0–w_0 basin map documents this basin dependence.

5 Branch feasibility and local stability

The interior branch exists only under explicit feasibility conditions:

$$c > 0, \quad r_u - ca_u > 0, \quad z^* > 0, \quad w^* > 0, \quad n^* > 0, \quad 0 < q^*(s) < 1.$$

Solving the branch equation for stress gives

$$s(q) = n^* z^* [b_u + \Delta b q] - m - \mu w^*.$$

Therefore the open stress interval for an interior branch is the interval between the values at $q = 0$ and $q = 1$. In the current parameterization this is approximately

$$s \in (-0.1131, 0.2324).$$

The target stresses used in the mechanism tests lie inside this interval. These conditions make the branch a conditional mathematical statement for the linear trade-off ODE rather than a purely numerical observation.

The endpoints of this analytic interior branch occur when $q^*(s)$ reaches the trait boundaries $q = 0$ or $q = 1$. They are feasibility or trait-boundary endpoints of the interior formula. We do not classify these endpoints as saddle-node, transcritical, or other bifurcations; a full boundary-equilibrium bifurcation analysis is outside the present scope.

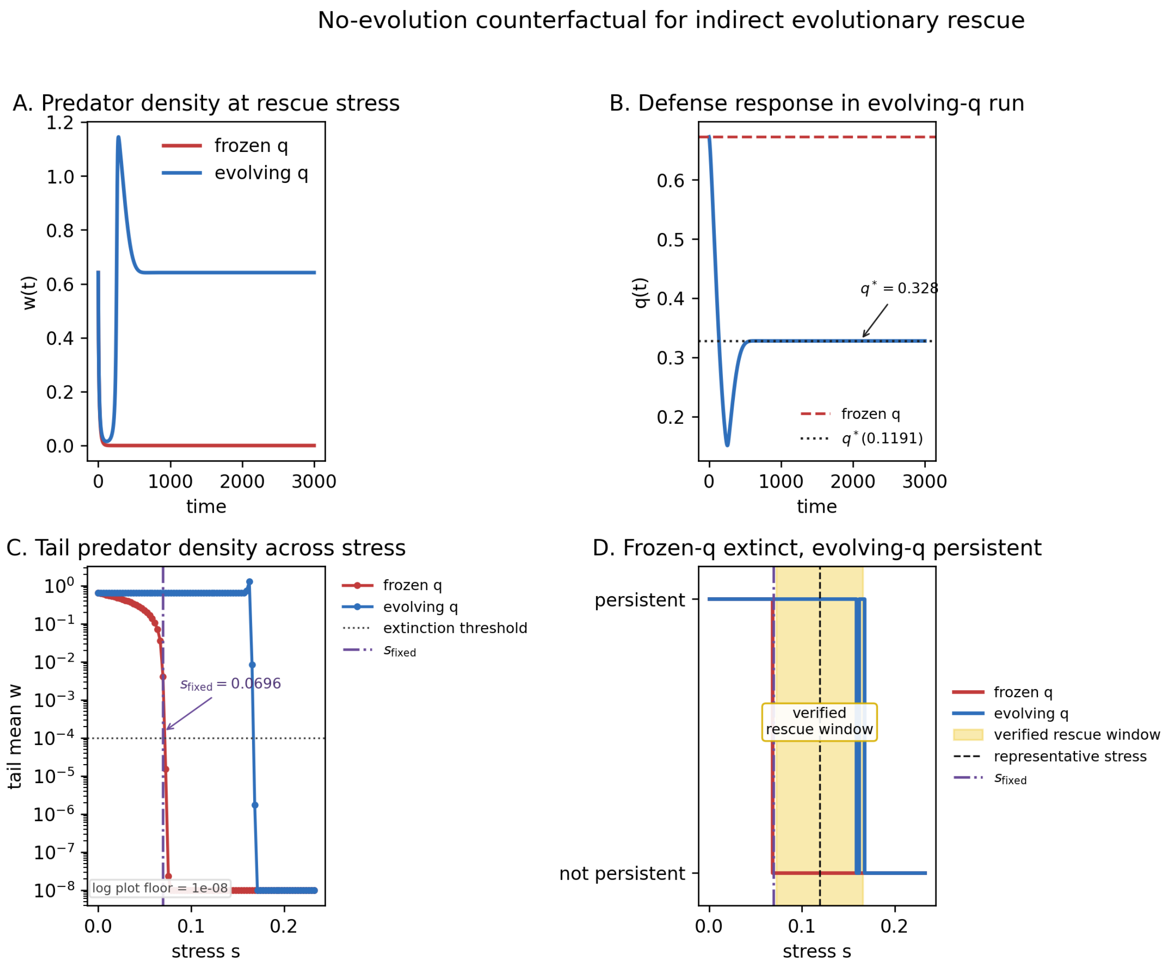

Figure 2: No-evolution counterfactual for the linear ODE. Panel A compares predator density at a representative rescue stress for evolving-q and frozen-q treatments. Panel B shows the defense response in the evolving-q run. Panel C plots tail mean predator density across stress for both treatments; values below 10^{-8} are shown at the plotting floor for log-scale visualization. Panel D highlights the finite stress window where frozen-q trajectories are extinct but evolving-q trajectories persist from the pre-stress initial condition.

Local stability of the homogeneous branch was evaluated with the analytic Jacobian of the ODE. For a three-dimensional system, the characteristic polynomial can be written

$$\lambda^3 + A_1\lambda^2 + A_2\lambda + A_3 = 0.$$

The cubic Routh–Hurwitz conditions for local stability are

$$A_1 > 0, \quad A_2 > 0, \quad A_3 > 0, \quad A_1 A_2 > A_3.$$

For the current parameterization, these inequalities hold at all target stresses tested along the branch and agree with direct eigenvalue stability. The smallest recorded Routh–Hurwitz margin $A_1A_2 - A_3$ across the target stresses was positive. This supports local stability of the homogeneous compensation branch in the tested stress interval. It does not prove global attraction or determine all basin boundaries.

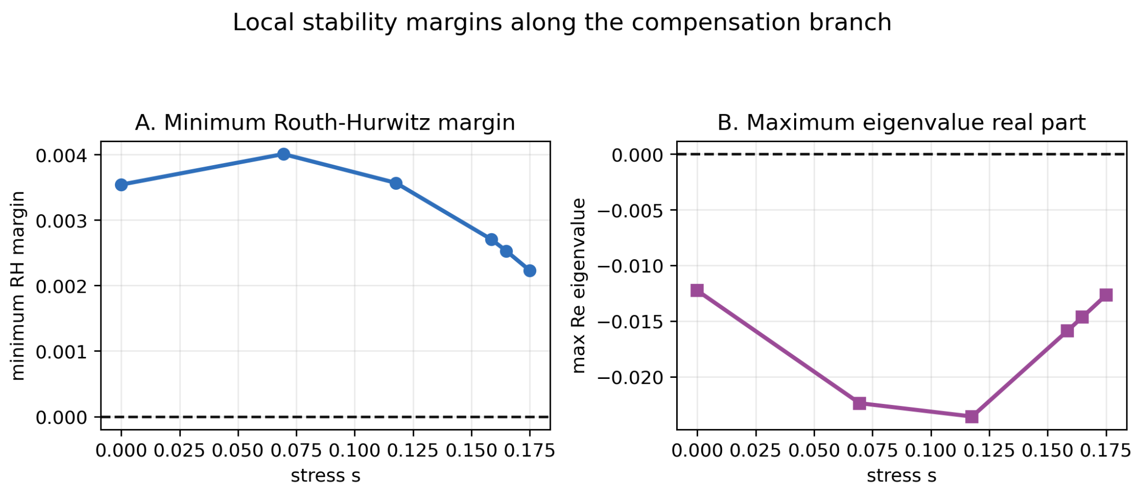

Figure 3: Local stability margins along the compensation branch. Panel A shows the minimum Routh–Hurwitz margin across A_1, A_2, A_3, and $A_1A_2 - A_3$. Panel B shows the maximum real part of the reaction-Jacobian eigenvalues.

6 Spatial PDE stability and finite-amplitude heterogeneity

A homogeneous ODE equilibrium is also a homogeneous PDE steady state because diffusion terms vanish for spatially constant fields. For Neumann spatial modes on a rectangular domain, perturbations evolve according to

$$J_F(U^*) - \lambda_{ij} D,$$

where $J_F(U^*)$ is the reaction Jacobian at the homogeneous branch state, $D = \mathrm{diag}(D_n, D_w, D_q)$, and

$$\lambda_{ij} = \left(\frac{i\pi}{L_x}\right)^2 + \left(\frac{j\pi}{L_y}\right)^2.$$

The zero mode is the ODE stability problem. Nonzero modes test whether infinitesimal spatial perturbations grow.

For the current diffusion coefficients, all tested nonzero Neumann modes with $i, j = 0, \ldots, 64$ were stable at the target branch stresses. A sampled continuous-λ scan of $\max \Re\mathrm{eig}(J_F(U^*) - \lambda D)$, using 600 equally spaced λ values over the corresponding mode-equivalent range, also found no positive growth. A controlled diffusion-ratio linear grid found no spatial instability in the tested ratios. Thus the PDE does not destabilize the homogeneous branch through the tested infinitesimal spatial modes and modal range. This is still a tested-range result, not an analytic proof over all spatial modes or all diffusion coefficients.

The q-diffusion closure matters for this spatial interpretation. In the modal matrix, the term $-\lambda D$ directly damps nonzero q-component perturbations through D_q. Therefore the negative spatial-pattern result is conditional on the phenomenological q-diffusion closure and tested diffusion values. A supplemental linear diagnostic varied $D_q \in \{0, 0.001, 0.005, 0.01, 0.05\}$ with $D_n = D_w = 0.01$; it found no positive sampled modal growth over the tested λ-range, including the $D_q = 0$ case, but this remains a linear diagnostic and not a nonlinear PDE robustness result.

Linear spatial stability is not enough to rule out finite-amplitude heterogeneous effects. Targeted finite-amplitude PDE tests therefore used localized predator patches, localized defense patches, sinusoidal perturbations, smooth random heterogeneity, and basin-boundary heterogeneity. These tests compared heterogeneous fields with matched homogeneous controls. Initial finite-horizon basin-label changes occurred for localized defense perturbations near basin-boundary mean states, but these changes were transient-label effects without persistent spatial patterning. Longer-horizon analysis showed that the basin-changing cases resolved to the matched homogeneous-control basin, with final spatial coefficients of variation below the persistence threshold.

The spatial interpretation is therefore specific: in the tested PDE, spatial heterogeneity can create transient excursions near basin boundaries, but it did not generate a persistent spatial-pattern-mediated rescue mechanism.

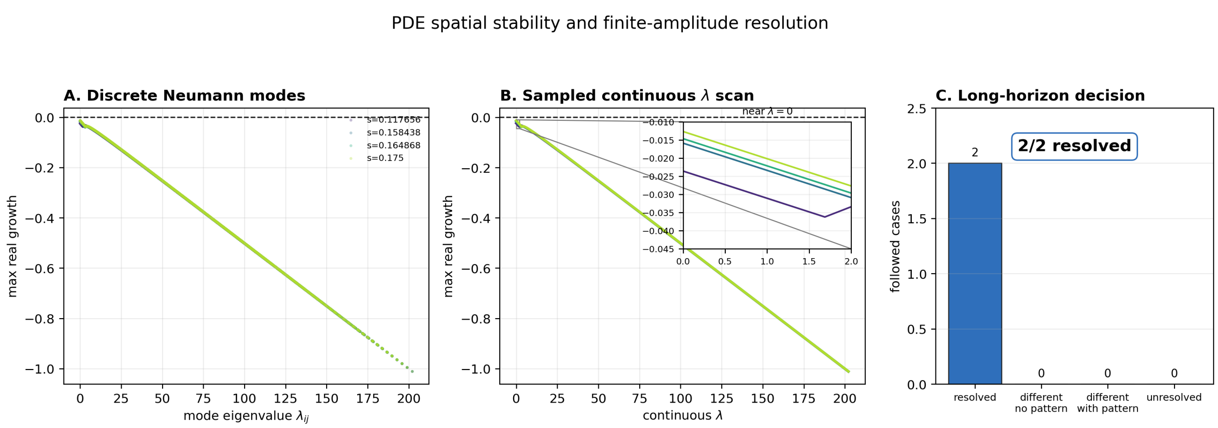

Figure 4: Spatial-mode and finite-amplitude PDE tests. Panel A plots maximum real growth rates for tested discrete Neumann modes, computed from $J_F(U^*) - \lambda_{ij} D$. Panel B shows the sampled continuous-λ scan over 600 equally spaced values in the mode-equivalent range corresponding to $i, j = 0, \ldots, 64$. Panel C summarizes the two followed finite-amplitude heterogeneous cases that changed finite-horizon basin labels; both resolved to their matched homogeneous controls without persistent spatial patterning.

7 Nonlinear trade-off extension

For nonlinear trade-offs, the branch is most naturally parameterized by q. When derivatives are defined and $a'(q) \neq 0$, the selection-gradient condition gives

$$c(q) = \frac{r'(q)}{a'(q)}.$$

The corresponding branch geometry is

$$z(q) = \frac{\xi}{r(q) - a(q)c(q)}, \quad w(q) = c(q)z(q), \quad n(q) = \kappa^{-1} - z(q) - w(q),$$

and predator balance maps the branch to stress:

$$s(q) = b(q)n(q)z(q) - m - \mu w(q).$$

The linear branch is recovered when all shape exponents are one.

The controlled nonlinear extension used $\gamma_r, \gamma_a, \gamma_b \in \{0.5, 1, 2\}$, representing concave, linear, and convex endpoint-preserving shapes. This grid is small and structured; it is not a global trade-off scan. Within that grid, the generalized formulation recovered the linear branch, and selected concave, convex, and mixed nonlinear shapes supported feasible locally stable compensation branches. The shape-grid summary recorded 11 robust compensation shapes, 11 partial compensation shapes, 4 no-compensation shapes, and 1 unresolved shape. Targeted PDE spatial-mode and non-homogeneous tests for selected stable nonlinear branches did not support persistent spatial-pattern-mediated rescue.

The nonlinear result suggests that compensation is not only a linear-trade-off artifact within this controlled shape grid. It also emphasizes parameter sensitivity: nonlinear shape can change branch feasibility, stability, and stress interval length. The no-compensation and unresolved nonlinear shapes are therefore informative failure modes, not exceptions to hide: in the existing diagnostics they correspond to missing feasible branches at the target stresses or to branches whose stability/classification was not sufficient to support the compensation claim, as detailed in the supplemental failure-mode table.

8 Biological interpretation

Higher predator mortality directly reduces predator persistence. In this model, prey defense evolution can compensate because mortality stress reduces predator pressure, and lower predator pressure favors lower prey defense. Lower defense increases predator conversion opportunity through $b(q)$, which can restore predator growth balance at a positive predator equilibrium. The frozen-q counterfactual verifies this as indirect evolutionary rescue on the finite tested stress grid where fixed prey defense leads to predator extinction but evolving prey defense preserves the predator.

The supported biological interpretation is a reaction-level compensation mechanism rather than a pattern-mediated one. Defended prey are less favorable to predators in the conversion term, so a shift toward lower defense makes prey more exploitable by the predator. Under the tested trade-off values, that evolutionary shift is sufficient to offset part of the imposed predator mortality stress. The predator persists indirectly because the prey population evolves toward a state that improves predator growth balance.

The spatial PDE does not create the rescue mechanism in the tested setup. Instead, the PDE preserves a homogeneous reaction-level branch that is locally stable to tested spatial modes. Spatial

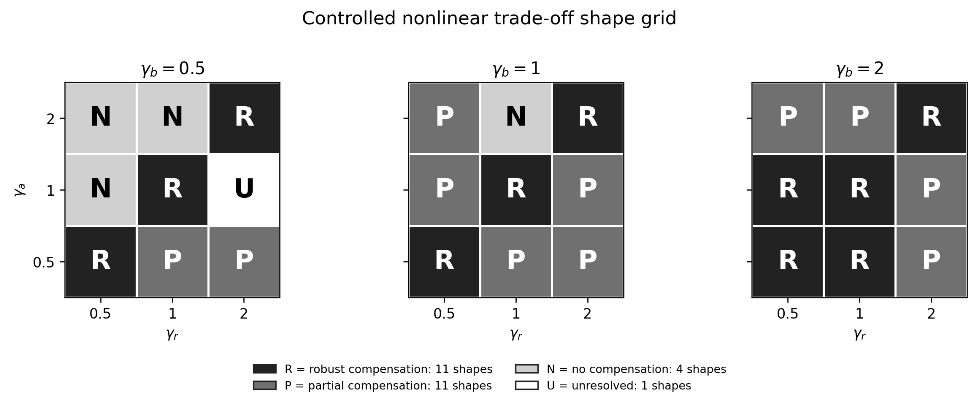

Figure 5: Controlled nonlinear trade-off extension summary. The figure shows the structured shape grid faceted by conversion-shape exponent γ_b, with growth and attack exponents on the axes. Cell categories match the numerical summary: 11 robust compensation shapes, 11 partial compensation shapes, 4 no-compensation shapes, and 1 unresolved shape.

stability matters biologically because it means small spatial variation around the homogeneous rescue state is damped rather than amplified under the current diffusion coefficients. The absence of persistent spatial patterns in targeted non-homogeneous tests further argues against a pattern-mediated mechanism here, while still allowing transient heterogeneity near basin boundaries.

Nonlinear trade-off shapes matter because biological costs of defense need not be linear. The controlled shape grid shows that compensation can persist under selected concave, convex, and mixed endpoint-preserving trade-offs, but it also shows that branch feasibility and stability are shape-dependent. Several assumptions remain stylized: the defense trait is represented by a single frequency q, trade-offs are low-dimensional functions, q-diffusion is phenomenological, diffusion is isotropic and constant, and environmental stress enters as an added predator mortality term. These choices make the mechanism interpretable but not biologically calibrated. Empirical prediction would require species-specific estimates of defense costs, predator conversion rates, movement scales, mortality stress, and the timescale of prey evolutionary response.

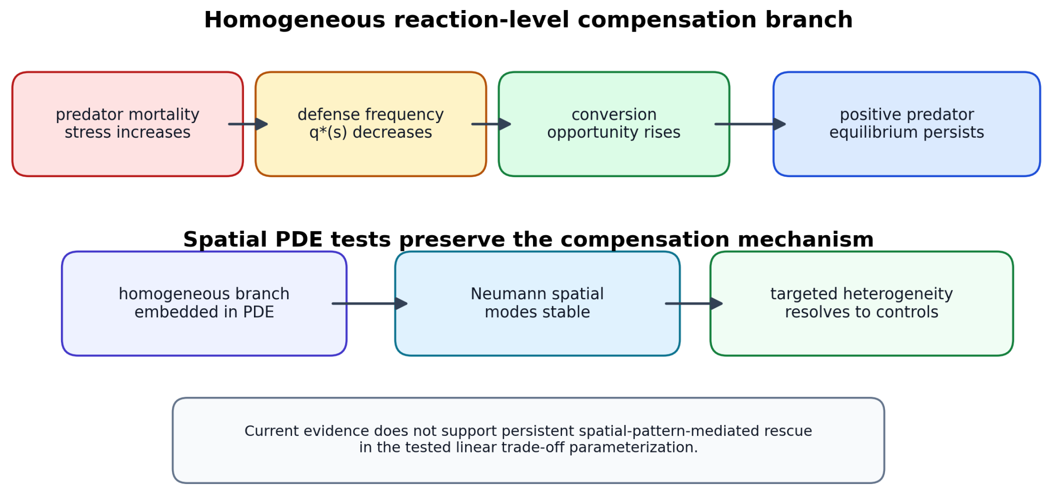

Figure 6: Final mechanism synthesis. The diagram summarizes the permitted mechanism claim: predator mortality stress shifts homogeneous eco-evolutionary balance, lower prey defense increases predator conversion opportunity, and the spatial PDE preserves this reaction-level mechanism in the tested setup. The figure does not imply spatial-pattern-mediated rescue or biological prediction without calibration.

9 Scope and limitations

The established result in this parameterization is that the well-mixed eco-evolutionary ODE has an interior compensation branch, that the branch is feasible over an explicit stress interval, that the branch is locally stable at the tested stresses by Routh–Hurwitz criteria, and that a frozen-q counterfactual verifies indirect evolutionary rescue over a finite stress window. For the spatial PDE, the supported result is that the same homogeneous branch is linearly stable over the tested Neumann modes and sampled continuous-λ modal range, and that targeted finite-amplitude heterogeneous perturbations do not produce persistent spatial-pattern-mediated rescue.

The controlled extension supports the presence of compensation branches beyond the linear trade-off form for selected concave, convex, and mixed endpoint-preserving shapes. This does not establish global robustness over all nonlinear trade-offs. The PDE tests are targeted rather than exhaustive. There is no global basin proof, no proof of global stability, no theorem for all heterogeneous initial conditions, and no full bifurcation analysis.

The branch equilibrium itself does not depend on the evolutionary-rate parameter ν, because the interior equilibrium condition sets the selection gradient to zero. Finite-time rescue from a specified initial condition can still depend on ν: if prey defense evolves too slowly, predators may decline before the defense shift restores predator growth balance. A supplemental finite-time diagnostic varied $\nu \in \{0.005, 0.01, 0.025, 0.05, 0.1\}$ from the pre-stress branch initial condition. At the representative rescue stress $s = 0.1191$, $\nu = 0.005$ lost the predator, $\nu = 0.01$ recovered mathematically after crossing below the extinction threshold, and $\nu \geq 0.025$ remained predator-persistent without threshold crossing under the tested horizon. This diagnostic does not alter the baseline $\nu = 0.05$ rescue-window claim.

The continuous-λ scan reduces dependence on the chosen domain for infinitesimal modes, because it samples a modal interval rather than only the discrete modes of a single rectangle.

The finite-amplitude PDE perturbation results remain conditional on the 20×20 Neumann domain. Larger domains, different boundary conditions, or different dispersal closures could allow longer-lived spatial refugia or different heterogeneous basin behavior; domain-size robustness is not claimed.

The model is stylized and not calibrated to empirical predator–prey data. Parameters are used to test mechanism, not to estimate a particular biological system. The manuscript does not claim that spatial structure generally amplifies or suppresses rescue, and it does not claim that all heterogeneous perturbations decay. The results should therefore be interpreted as a mechanistic case study: they support homogeneous reaction-level compensation in the tested Roy-style model, not a universal claim about spatial evolutionary rescue.

10 Conclusions

1. The tested ODE supports indirect evolutionary rescue through prey defense evolution over the verified frozen-q counterfactual window $0.0726 \leq s \leq 0.1656$ on the stress grid.
2. The mechanism is a homogeneous compensation branch.
3. Branch existence follows from explicit feasibility conditions.
4. The current branch is locally stable by Routh–Hurwitz criteria.
5. The spatial PDE branch is linearly stable over the tested Neumann spatial-mode range and sampled continuous-λ modal range.
6. Targeted non-homogeneous perturbations do not produce persistent spatial-pattern-mediated rescue.
7. Controlled nonlinear trade-off tests show that the mechanism extends beyond the linear case but remains parameter-sensitive.

The current evidence supports homogeneous reaction-level compensation, not spatial-pattern-mediated rescue, in the tested model.

Data and code availability

All scripts, CSV outputs, figures, and LaTeX sources are maintained in the repository at https://github.com/Thithilia/Mathbio. The external-review snapshot for this manuscript package is identified by the tag `external-review-polish-2026-05-31`. The manuscript is based on the CSV and figure outputs generated by the numbered analysis scripts, including ODE no-evolution counterfactual outputs, linear modal-scan outputs, and manuscript value/DOI audit tables; these analyses do not add PDE simulations or change model equations. No release DOI exists yet; the repository should be archived before journal submission.

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